

In this project, a Multi-Agent Framework designed for **autonomous, contactless clinical triage**. The system aims to:

1. **Extract** accurate physiological biomarkers (Heart and Respiratory Rates) from standard RGB video feeds using rPPG(Remote Photoplethysmography) technology.
2. **Analyze** these biomarkers through a **Multi-Agent system** where specialized AI agents (Perception, Diagnostic, and Coordinator) interact to "reason" about patient severity.
3. **Prioritize** patients dynamically on a centralized **Triage Dashboard**, specifically flagging critical conditions like Compensated Shock to ensure immediate clinical intervention without the need for expensive hardware or physical contact.

In modern healthcare settings, particularly in emergency departments and high-traffic outpatient clinics, the traditional triage process is heavily reliant on manual assessments by medical personnel. This dependency introduces three critical vulnerabilities:

- **Subjectivity and Human Error:** Manual vital sign assessment is periodic and prone to fatigue-induced errors, which can lead to the misclassification of patient urgency.
- **Cross-Infection Risks:** Contact-based monitoring (using pulse oximeters or BP cuffs) increases the risk of pathogen transmission between patients and healthcare workers.
- **Delayed Detection:** Life-threatening conditions such as **Compensated Shock**—characterized by simultaneous Tachycardia (high heart rate) and Tachypnea (high respiratory rate)—often go undetected in crowded waiting areas until the patient reaches a state of decompensation.

rPPG Technology: Reference the shift from contact-based PPG to **Eulerian Video Magnification** and **Green-channel analysis** . This proves the feasibility of getting a pulse from a camera.

Background : **Emergency Severity Index (ESI)** is the gold standard for triage, but manual assessment is slow. **Compensated Shock** (high heart rate + high respiratory rate) is often missed in crowded waiting rooms

Architecture “: Define the three agents:

- **Perception Agent:** Extracts raw data from video.
- **Diagnostic Agent:** Calculates BPM and checks against medical rules.
- **Coordinator Agent:** Ranks patients on a Triage Dashboard.

Dataset Name	Content	Why it's useful
UBFC-rPPG	Videos of people playing a math game.	Includes heart rate spikes due to stress (perfect for triage).
PURE	Videos with different head movements.	Tests if the AI can handle moving patients.
MMSE-HR	High-resolution videos with emotion.	Good for testing stress and anxiety analytics.

Most modern hospitals use a 5-level scale to categorize patients during triage:

Level	Category	Urgency	Example Condition
1	Resuscitation	Immediate (Life-saving)	Cardiac arrest, severe trauma.
2	Emergent	Within 10–15 minutes	Chest pain, stroke, severe asthma.
3	Urgent	Within 30–60 minutes	Abdominal pain, moderate high fever.
4	Semi-Urgent	Within 1–2 hours	Sprained ankle, earache, small cuts.
5	Non-Urgent	Can wait (Routine)	Cold symptoms, prescription refills.

Development	Google Colab	The Free Tier provides GPUs for model training. It does not require a card; only a standard Gmail account.
Vision AI	MediaPipe	An open-source library from Google. There is no usage cost or API fee.

Agentic Brain	Groq Cloud API	They offer a Free Tier for Llama-3 models. It is limited by speed (RPM), but plenty for a student project.
Agent Framework	CrewAI / LangChain	These are Open Source Python libraries. You install them via pip, and they are free forever.
Dashboard UI	Streamlit Cloud	You can host public apps for Free as long as the code is on GitHub.
Medical Data	UBFC-rPPG	A research-grade dataset available for Free (downloadable from Kaggle or University sites).

Compensated shock" is the stage where the body is working hard to maintain blood pressure despite poor circulation. In an engineering context, this is a **multivariate correlation problem**. The students can test this using the following "Triage Rule Table":

Metric	Normal Range	Compensated Shock (Tachypnea)	Why this happens?
Respiration Rate (RR)	12–20 bpm	> 20 bpm	To blow off \$CO_2\$ and fix metabolic acidosis.
Heart Rate (HR)	60–100 bpm	> 100 bpm (Tachycardia)	To pump blood faster to vital organs.
Pulse Pressure	~40 mmHg	Narrowing (< 25 mmHg)	Due to systemic vasoconstriction.

Component	Free Tool	Agentic Function
Orchestration	LangChain	Manages the "thoughts" and "actions" of the agents.

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Logic/Reasoning	Ollama or Groq Cloud	Provides a free API for a Large Language Model (LLM) to "think" about the vitals.
Vision Extraction	MediaPipe	The tool used by the Vision Agent to see the patient.
Dashboard	Streamlit	The interface where the agents display their "Reasoning Logs."

In a traditional setup, the code just follows a fixed path (Video → Heart Rate → Display). In an **Agentic setup**, the AI acts as a "Medical Coordinator." It observes the data, reasons about the severity, and decides which "tool" or "action" to take next.

Agent 1: The Vision Observer (Perception Agent)

- **Role:** Uses the webcam to detect the patient and extract the rPPG signal.
- **Intelligence:** If the lighting is too dark or the patient is moving too much, the agent doesn't just fail—it **reasons**. It tells the user: *"Please stay still and increase room lighting for an accurate reading."*

Agent 2: The Diagnostic Analyst (Reasoning Agent)

- **Role:** Receives the heart rate and respiratory data.
- **Intelligence:** It compares the current vitals against a medical knowledge base (like a digital version of the "WHO Triage Guidelines").
- **Action:** If it sees a Heart Rate of 130 BPM *and* a high Respiration Rate, it concludes: *"This patient is in Compensated Shock."*

Agent 3: The Triage Coordinator (Action Agent)

- **Role:** Manages the dashboard and alerts.
- **Intelligence:** It looks at all patients in the "waiting room" and **re-prioritizes the list** dynamically.
- **Action:** It sends an automated notification to the doctor's phone: *"Emergency: Patient in Seat 4 requires immediate attention due to Tachycardia."*

The alternative to a live camera is a **Digital Video Pipeline** using pre-recorded research datasets.

1. The Alternative Workflow: "Video-as-a-Stream"

Instead of the `cv2.VideoCapture(0)` command (which opens a webcam), the students will use: `cv2.VideoCapture("path_to_dataset_video.mp4")`.

The system will treat each frame of the recorded video as if it were arriving "live" from a camera

Procedure:

Step 1: Environment Setup & Data Acquisition (The Foundation)

- **The Action:** Students must set up their coding environment and download the "Patient Data."
- you won't be using a live webcam for development. You will use the **UBFC-rPPG dataset**. This is your 'Virtual Patient.' You need to set up **Python** and install libraries like **MediaPipe** and **OpenCV**. By the end of this step, you should be able to load a video file and have the computer track a face frame-by-frame."

Step 2: The Perception Layer (Signal Extraction)

- **The Action:** Isolating the skin and finding the "pulse" in the video.
- Next, you build the **Perception Agent**. You will use MediaPipe to 'crop' the forehead and cheeks. You are looking for the **Green Color Channel**, which carries the strongest heartbeat signal. You will then apply a **Bandpass Filter** to remove 'noise' like shadows or room lighting. Your goal here is to see a wavy line (a pulse) on your screen."

Step 3: Constructing the Multi-Agent Framework (The Brain)

- **The Action:** Connecting the AI Agents using CrewAI.
- **The Narrative:** "Now we add the intelligence. You will create three 'Agents' that talk to each other.
 1. The **Perception Agent** calculates the Heart Rate from the wave.
 2. The **Diagnostic Agent** takes that number and asks the **Groq Llama-3 API**: 'If the heart rate is 120 and the patient is breathing fast, what is the risk?'
 3. The **Coordinator Agent** then ranks this patient against others."

Step 4: Building the Triage Dashboard (The Interface)

- **The Action:** Creating the UI using Streamlit.
- You will use **Streamlit** to build a dashboard. It should show the video file playing, the live heartbeat graph moving, and a 'Triage List' that automatically moves the sickest patients to the top with a red 'Critical' alert."

Step 5: Validation & Error Analysis (The Final Proof)

- **The Action:** Comparing the AI results to the real medical sensors in the dataset.
- Finally, you must prove your AI is accurate. You will compare the heart rate your AI calculated against the **Ground Truth** sensor data provided in the dataset. You will calculate the **Mean Absolute Error (MAE)**. If your error is minimal, you have successfully built a clinical-grade tool .

Detecting heart rate through the face is a scientific process called **Remote Photoplethysmography (rPPG)**. It relies on the fact that your skin acts like a mirror that reflects the internal "pulses" of your blood.

Here is the simple, step-by-step explanation you can give your students:

1. The "Invisible" Color Change

Every time the heart beats, it pumps a fresh wave of blood through the arteries. This causes the tiny blood vessels (capillaries) in the face to expand slightly.

- **The Science:** When these vessels expand, they absorb more light. When they contract between beats, they reflect more light.
- **The Camera's Role:** While the human eye cannot see these tiny shifts, a digital camera "sees" them as minute changes in pixel intensity (color).

2. Why the "Green Channel" is Key

A digital camera records in Red, Green, and Blue (RGB). But focus on the **Green channel**.

- **Hemoglobin Absorption:** Oxygenated blood (hemoglobin) absorbs green light much better than it absorbs red or blue light.
- **Strongest Signal:** Because of this high absorption, the "Green" signal from the face has the most contrast, making it the easiest to turn into a heart rate wave.

3. The Signal Extraction Process

1. **Face Tracking:** Use **MediaPipe** to find the face and lock onto the **Forehead** and **Cheeks**. These areas have the thinnest skin and the most blood vessels.

2. **Spatial Averaging:** Take all the green pixels in those areas and find their average value for every single frame. This turns a video into a 1D "signal."
3. **Denoising (Filtering):** A raw signal is messy because of room lights or head movements. Students apply a **Bandpass Filter** (between 0.7 Hz and 4 Hz) to remove any data that isn't in the range of a human heartbeat.
4. **Frequency Analysis (FFT):** Finally, they use a **Fast Fourier Transform (FFT)**. This tool finds the most "dominant" rhythm in that wave, which is the Heart Rate.

Think of the forehead as a rhythmic light bulb. Every time your heart beats, the bulb dims just a tiny bit because the blood is absorbing the light. Our AI Agent's job is to ignore the background noise and count how many times that 'bulb' dims per minute. That count is your Heart Rate

<https://www.youtube.com/watch?v=kJkk1AozxFw>